

Math142 - Practice Problems for Exam 2

- Determine whether the sequence is convergent or divergent. If it is convergent, find its limit.
 - $a_n = \frac{n \sin n}{n^2+1}$
 - $a_n = (1 + \frac{3}{n})^n$
- Determine whether the series is convergent or divergent.
 - $\sum_{n=1}^{\infty} \ln(\frac{n}{3n+1})$
 - $\sum_{n=1}^{\infty} \frac{n}{\sqrt{n^5+n+1}}$
 - $\sum_{n=2}^{\infty} \frac{1}{n\sqrt{\ln n}}$
 - $\sum_{n=1}^{\infty} \frac{n^{2n}}{(1+2n^2)^n}$
 - $\sum_{n=1}^{\infty} \frac{\cos^2 n}{n^{4/3}}$
 - $\sum_{n=1}^{\infty} n e^{-n}$
 - $\sum_{n=1}^{\infty} \frac{3^n n^2}{n!}$
 - $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n^n}{n!}$
- Determine whether the series is conditionally convergent, absolutely convergent or divergent.
 - $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n^3}$
 - $\sum_{n=2}^{\infty} (-1)^n \sin(\frac{\pi}{n})$
 - $\sum_{n=1}^{\infty} \frac{\sin 2n}{1+2^n}$
 - $\sum_{n=1}^{\infty} (-1)^n \frac{\ln n}{\sqrt{n}}$
- Find the radius of convergence and interval of convergence of the series.
 - $\sum_{n=1}^{\infty} \sqrt{n} x^n$
 - $\sum_{n=1}^{\infty} (-1)^n \frac{x^n}{n^2 2^n}$
 - $\sum_{n=1}^{\infty} \frac{3^n (x-3)^n}{\sqrt{n+3}}$
- Find the Maclaurin series (power series centered at 0) and its radius of convergence.
 - $f(x) = \frac{x^2}{1+x}$
 - $f(x) = x e^{2x}$
 - $f(x) = \sin(x^2)$.